

习题 2.1. 对方程组

$$\begin{cases} x_1 + 2x_2 - x_3 - x_4 = 0, \\ x_1 + 2x_2 + x_4 = 4, \\ -x_1 - 2x_2 + 2x_3 + 4x_4 = 5, \end{cases}$$

求导出组的一个基础解系、一个特解，并写出通解形式。

习题 2.2. 求线性方程组

$$\begin{pmatrix} \lambda-2 & 2 & 0 \\ 2 & \lambda-3 & 2 \\ 0 & 2 & \lambda-4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}$$

的解集与参数 λ 的关系。

1.

$$\begin{bmatrix} 1 & 2 & -1 & -1 & 0 \\ 1 & 2 & 0 & 1 & 4 \\ -1 & -2 & 2 & 4 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 & 4 \\ 0 & 0 & -1 & -2 & -4 \\ 0 & 0 & 2 & 5 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 & 4 \\ 0 & 0 & 1 & 2 & 4 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\therefore \text{一个特解为: } \gamma_0 = (1, 2, 1)^T$$

$$\text{导出组解为 } \begin{cases} x_1 = -2x_2 \\ x_3 = x_4 = 0 \end{cases}$$

$$x_2 \text{ 为自由变量, 基础解系 } W_0 = \{k\eta_1, k \in K\}, \eta_1 = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore \text{通解为: } W = \{\gamma_0 + k_1\eta_1, k_1 \in K\}$$

2.

$$\begin{vmatrix} \lambda-2 & 2 & 0 \\ 2 & \lambda-3 & 2 \\ 0 & 2 & \lambda-4 \end{vmatrix} = (\lambda-2)[\lambda^2-7\lambda+8] - 2(2\lambda-8) = \lambda^3-7\lambda^2+8\lambda-2\lambda^2+14\lambda-16+2\lambda+16 \\ = \lambda^3-9\lambda^2+18\lambda \\ = \lambda(\lambda^2-9\lambda+18) \\ = \lambda(\lambda-3)(\lambda-6)$$

① 若 $\lambda \neq 0$ 且 $\lambda \neq 3, \lambda \neq 6$, 则 $\text{rank}(A) = 3 = \text{rank}(\tilde{A})$,

且 $\text{rank} = \text{未知数个数}$, 必定有唯一解

② $\lambda = 0$ 时,

$$\begin{vmatrix} -2 & 2 \\ 2 & -3 \end{vmatrix} = 6-4=2 \neq 0, \therefore \text{rank}(A)=2.$$

$$\begin{vmatrix} 2 & 0 & 0 \\ -3 & 2 & 1 \\ 2 & -4 & -3 \end{vmatrix} = 2 \cdot \begin{vmatrix} 2 & 1 \\ -4 & -3 \end{vmatrix} = 2 \cdot (-6+4) = -4 \neq 0,$$

$\therefore \text{rank}(\tilde{A}) \neq \text{rank}(A)$, 无解

③ $\lambda = 3$ 时,

$$\begin{vmatrix} 0 & 2 \\ 2 & -1 \end{vmatrix} \neq 0, \text{rank}(A)=2, \begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 2 & -1 & -3 \end{vmatrix} = 2 \cdot \begin{vmatrix} 2 & 1 \\ -1 & -3 \end{vmatrix} \neq 0$$

$3 = \text{rank}(\tilde{A}) \neq \text{rank}(A) = 2$, 无解

④ $\lambda = 6$ 时,

$$\begin{vmatrix} 4 & 2 \\ 2 & 3 \end{vmatrix} = 12-4=8 \neq 0, \therefore \text{rank}(A)=2$$

$$\begin{vmatrix} 2 & 0 & 0 \\ 3 & 2 & 1 \\ 2 & 2 & -3 \end{vmatrix} = 2 \cdot \begin{vmatrix} 2 & 1 \\ 2 & -3 \end{vmatrix} \neq 0, \therefore \text{rank}(\tilde{A}) \neq \text{rank}(A)$$

$\therefore \lambda = 0/3/6$ 时, 无解

习题 2.3. 求行列式

$$\begin{vmatrix} 5 & 7 & -2 & 4 \\ 1 & 1 & 0 & -5 \\ -1 & 3 & 1 & 3 \\ 2 & -4 & -1 & -3 \end{vmatrix}$$

原式=

$$\begin{vmatrix} 1 & 1 & 0 & -5 \\ -1 & 3 & 1 & 3 \\ 5 & 7 & -2 & 4 \\ 2 & -4 & -1 & -3 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 0 & -5 \\ 0 & 4 & 1 & -2 \\ 0 & 2 & -2 & 29 \\ 0 & -6 & -1 & 7 \end{vmatrix} = \begin{vmatrix} 4 & 1 & -2 \\ 2 & -2 & 29 \\ -6 & -1 & 7 \end{vmatrix} = - \begin{vmatrix} 2 & -2 & 29 \\ 0 & 5 & -60 \\ 0 & -7 & 94 \end{vmatrix}$$

$$= -10 \cdot \begin{vmatrix} 1 & -12 \\ -7 & 94 \end{vmatrix} = -10 \cdot (94 - 84) = -100$$

习题 2.4. 设有向量组

$$\alpha_1 = (2, 3, 4, 7), \alpha_2 = (5, -1, 3, 2), \alpha_3 = (-3, 4, 1, 5), \alpha_4 = (0, -1, 7, 2), \alpha_5 = (6, 2, 1, 5),$$

证明 α_1, α_2 线性无关, 并求出所有包含 α_1, α_2 的极大无关组。

$$\therefore \begin{vmatrix} 2 & 5 \\ 3 & -1 \end{vmatrix} = -2 - 15 = -17 \neq 0$$

$$\frac{17}{119}$$

$\therefore (2, 3)$ 与 $(5, -1)$ 线性无关, 而 α_1, α_2 是延伸组, 也线性无关

$$\begin{bmatrix} 2 & 3 & 4 & 7 \\ 5 & -1 & 3 & 2 \\ -3 & 4 & 1 & 5 \\ 0 & -1 & 7 & 2 \\ 6 & 2 & 1 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 7 \\ -3 & 4 & 1 & 5 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 7 & 2 \\ 0 & 10 & 3 & 15 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 7 \\ 0 & \frac{17}{2} & -5 & \frac{31}{2} \\ 0 & -1 & 7 & 2 \\ 0 & 10 & 3 & 15 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 & 7 \\ 0 & -1 & 7 & 2 \\ 0 & 0 & \frac{109}{2} & \frac{65}{2} \\ 0 & 0 & 73 & 35 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$\therefore \text{rank}(\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5) = 4$ 而 $\alpha_1, \alpha_2, \alpha_3$ 线性相关

$\therefore \alpha_1, \alpha_2, \alpha_4, \alpha_5$ 是一个极大线性无关组.

习题 2.5. 写出矩阵

$$A = \begin{pmatrix} 3 & 2 & 0 & 5 & 0 \\ 3 & -2 & 3 & 6 & -1 \\ 1 & 6 & -4 & -1 & 4 \\ 2 & 0 & 1 & 5 & -3 \end{pmatrix}$$

的一个子式，并证明其为 A 的一个最高阶非零子式。

$$\begin{bmatrix} 1 & 6 & -4 & -1 & 4 \\ 2 & 0 & 1 & 5 & -3 \\ 3 & 2 & 0 & 5 & 0 \\ 3 & -2 & 3 & 6 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & -4 & -1 & 4 \\ 0 & -12 & 9 & 7 & -11 \\ 0 & -16 & 12 & 8 & -12 \\ 0 & -4 & 3 & 1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & -4 & -1 & 4 \\ 0 & -12 & 9 & 7 & -11 \\ 0 & -4 & 3 & 2 & -3 \\ 0 & -4 & 3 & 1 & -1 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 6 & -4 & -1 & 4 \\ 0 & -4 & 3 & 1 & -1 \\ 0 & 0 & 0 & 4 & -8 \\ 0 & 0 & 0 & 1 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 6 & -4 & -1 & 4 \\ 0 & -4 & 3 & 1 & -1 \\ 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & -2 \end{bmatrix}$$

$\therefore \text{rank}(A) = 4$ 不为 0 子式最高阶数为 4

且后四列组成子式 $\begin{vmatrix} 2 & 0 & 5 & 0 \\ -2 & 3 & 6 & -1 \\ 6 & -4 & -1 & 4 \\ 0 & 1 & 5 & -3 \end{vmatrix}$ 是一个非零子式

习题 2.6. 设 A 为已知的 n 阶方阵，满足 $\det(A) = 0$ 且 $A_{23} \neq 0$ ，求以 A 为系数矩阵的齐次线性方程组的解集。

习题 2.7. 已知向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关，其中 $n > 100$ 。问：

$$\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \dots, \alpha_{n-1} + \alpha_n, \alpha_n + \alpha_1$$

是否线性无关？请证明你的结论。

2.6 $\because \det(A) = 0$ 且 $A_{23} \neq 0$ $\therefore \text{rank}(A) = n-1$ ，基础解系维数为 1

设 $\eta = (A_{21}, A_{22}, \dots, A_{2n})'$ ， $\because A_{23} \neq 0$ $\therefore \eta \neq 0$ ，记 $A = (a_{ij})_{n \times n}$

而当 $i \neq 2$ 时， $\sum_{k=1}^n a_{ik} \cdot A_{2k} = 0$

$i=2$ 时 $\sum_{k=1}^n a_{2k} \cdot A_{2k} = |A| = 0$

$\therefore \sum_{k=1}^n a_{ik} \cdot A_{2k} = 0, i=1, 2, \dots, n$

$\therefore \eta$ 是一个解

$\therefore$ 解集 $W = \{k\eta, k \in K\}$

$$2-7. \quad k_1(d_1+d_2) + k_2(d_2+d_3) + \dots + k_n(d_n+d_1)$$

$$= (k_1+k_n)d_1 + (k_1+k_2)d_2 + (k_2+k_3)d_3 + \dots + (k_{n-2}+k_{n-1})d_{n-1} + (k_{n-1}+k_n)d_n = 0$$

$\therefore d_1, d_2, \dots, d_n$ 线性无关

$$\therefore \begin{cases} k_1+k_n=0 \\ k_1+k_2=0 \\ \vdots \\ k_{n-1}+k_n=0 \end{cases}, \text{ 是一个 } n \text{ 元齐次方程}$$

$$\begin{vmatrix} 1 & 0 & 0 & \dots & 0 & 1 \\ 1 & 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & \dots & 0 & 0 \\ 1 & 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 1 \end{vmatrix} - \begin{vmatrix} 0 & 0 & \dots & 0 & 1 \\ 1 & 1 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 1 \end{vmatrix}$$

$$= 1 - (-1)^n \begin{vmatrix} 1 & 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & 1 & \dots & 0 & 0 \\ 0 & 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 1 \\ 0 & 0 & 0 & \dots & 0 & 1 \end{vmatrix}$$

$$= 1 - (-1)^n = 1 + (-1)^{n-1}$$

$\therefore$ 当 n 为奇数时, 上方程只有零解, 线性无关

n 为偶数时, 有非零解, 线性相关

习题 2.8. 设矩阵

$$A = \begin{pmatrix} 3 & a & -1 & 4 \\ 1 & -1 & 1 & 1 \\ 5 & 3 & b & 6 \end{pmatrix},$$

判断是否存在 a, b 的取值使得 $\text{rank}(A) = 2$, 并说明理由。

$$\begin{bmatrix} 3 & a & -1 & 4 \\ 1 & -1 & 1 & 1 \\ 5 & 3 & b & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & a+3 & -4 & 1 \\ 0 & 8 & b-5 & 1 \end{bmatrix}$$

易知 $\begin{vmatrix} -1 & 4 \\ 1 & 1 \end{vmatrix} = -5 \neq 0$, $\therefore \text{rank}(A) \geq 2$, 而 A 只有 3 行, $\text{rank}(A) \leq 3$

$\therefore \text{rank}(A) = 2/3$

只要令两个 3 阶子式均为 0

$$\begin{vmatrix} 1 & -1 & 1 \\ 0 & a+3 & -4 \\ 0 & 8 & b-5 \end{vmatrix} = (a+3)(b-5) + 32 = ab + 3b - 5a + 17$$

$$\begin{vmatrix} -1 & 1 & 1 \\ a+3 & -4 & 1 \\ 8 & b-5 & 1 \end{vmatrix} = \begin{vmatrix} -1 & 1 & 1 \\ a+4 & -5 & 0 \\ 9 & b-6 & 0 \end{vmatrix} = \begin{vmatrix} a+4 & -5 \\ 9 & b-6 \end{vmatrix} = (a+4)(b-6) + 45 = ab + 4b - 6a + 21$$

$$\therefore \begin{cases} ab + 3b - 5a + 17 = 0 \\ ab + 4b - 6a + 21 = 0 \end{cases}$$

$$\therefore b - a + 4 = 0. \text{ 即: } b = a - 4$$

$$\therefore \begin{cases} a^2 - 4a + 3a - 12 - 5a + 17 = a^2 - 6a + 5 = 0 \\ a^2 - 4a + 4a - 16 - 6a + 21 = a^2 - 6a + 5 = 0 \end{cases}$$

$$\therefore \begin{cases} a=1 \\ b=-3 \end{cases} \text{ 或 } \begin{cases} a=5 \\ b=1 \end{cases} \text{ 时, } \text{rank}(A) = 2.$$

习题 2.9. 计算行列式:

$$\begin{vmatrix} x_1 + a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & x_2 + a_2 b_2 & \cdots & a_2 b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_n b_1 & a_n b_2 & \cdots & x_n + a_n b_n \end{vmatrix}.$$

$$\text{令 } e_i = (0, 0, \dots, \underset{\substack{\downarrow \\ \text{第 } i \text{ 列}}}{x_i}, 0, \dots, 0)'$$

$$\beta = (a_1, a_2, \dots, a_n)'$$

$$\therefore |A| = \det(e_1 + b_1 \beta, e_2 + b_2 \beta, \dots, e_n + b_n \beta), \text{ 对行列式作拆分}$$

由于两列成比例值为 0

$$\therefore |A| = \det(e_1, e_2, \dots, e_n) + \sum_{i=1}^n (e_1, e_2, \dots, e_{i-1}, b_i \beta, e_{i+1}, \dots, e_n)$$

$$\text{而 } \begin{vmatrix} x_1 & a_1 b_i \\ & x_2 & a_2 b_i \\ & & \ddots & \vdots \\ & & & a_i b_i & \ddots \\ & & & \vdots & & x_n \\ & & & a_n b_i & & \end{vmatrix} = x_1 x_2 \cdots x_n \cdot a_i b_i = \left(\prod_{i=1}^n x_i \right) \cdot \frac{a_i b_i}{x_i}$$

$$\therefore \text{原式} = \sum_{j=1}^n \left[\left(\prod_{i=1}^n x_i \right) \left(\frac{a_j \cdot b_j}{x_j} + 1 \right) \right]$$

Ex 1.

(1) 设 $\dim(V_1) = n_1, \dim(V_2) = n_2, \dim(V_1 \cap V_2) = m$

设 $V = V_1 \cap V_2$ 的一个基 $\beta_1, \beta_2, \dots, \beta_m$

则它可拓展为 V_1, V_2 的一个基, 分别记作

$$\beta_1, \beta_2, \dots, \beta_m, d_1, \dots, d_{n_1-m}$$

$$\beta_1, \beta_2, \dots, \beta_m, \gamma_1, \dots, \gamma_{n_2-m}$$

则 $V_1 + V_2$ 可被 $\beta_1, \beta_2, \dots, \beta_m, d_1, d_2, \dots, d_{n_1-m}, \gamma_1, \gamma_2, \dots, \gamma_{n_2-m}$ 线性表出

只要证它们线性无关, 则有

$$\dim(V_1 + V_2) = n_1 + n_2 - m = \dim(V_1) + \dim(V_2) - \dim(V_1 \cap V_2)$$

设 $l_1\beta_1 + l_2\beta_2 + \dots + l_m\beta_m + b_1d_1 + \dots + b_{n_1-m}d_{n_1-m} + c_1\gamma_1 + \dots + c_{n_2-m}\gamma_{n_2-m} = 0$

$$\therefore l_1\beta_1 + l_2\beta_2 + \dots + l_m\beta_m + b_1d_1 + \dots + b_{n_1-m}d_{n_1-m} = -c_1\gamma_1 - \dots - c_{n_2-m}\gamma_{n_2-m}$$

等式左边是 V_1 中的一个元素, 从而等式右侧在 $V_1 \cap V_2 = V$ 中

即存在 $r_1, r_2, \dots, r_m$ 使 $-c_1\gamma_1 - c_2\gamma_2 - \dots - c_{n_2-m}\gamma_{n_2-m} = r_1\beta_1 + \dots + r_m\beta_m$

即: $c_1\gamma_1 + c_2\gamma_2 + \dots + c_{n_2-m}\gamma_{n_2-m} + r_1\beta_1 + \dots + r_m\beta_m = 0$

而它们是 V_2 中线性无关的

$$\therefore c_1 = c_2 = \dots = c_{n_2-m} = 0$$

同理知 $l_1 = l_2 = \dots = l_m = b_1 = \dots = b_{n_1-m} = 0$

$\therefore$ 它们线性无关.

(2) 设 A, B 的解空间分别为 W_A, W_B

$$\begin{aligned} \therefore \dim(W_A \cap W_B) &= -\dim(W_A + W_B) + \dim(W_A) + \dim(W_B) \\ &= 2n - \text{rank}(A) - \text{rank}(B) - \dim(W_A + W_B) \\ &> 0 \end{aligned}$$

(3) $\because V_1 \subset V, \therefore V_1 \cap V = V_1$

$$\begin{aligned} \dim(V_1 + V) &= \dim(V_1) + \dim(V) - \dim(V \cap V_1) \\ &= \dim(V) \end{aligned}$$

$$(4). \quad AB = (A\beta_1, A\beta_2, \dots, A\beta_m)$$